Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahajbhadja | sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building **TypeScript-based user workflows**, **client-facing web platforms**, and API-driven systems, with experience integrating microservices, improving usability, and supporting complex data-heavy interfaces.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*
- Designed and optimized data-heavy user workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and backend service improvements.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, improving workflow reliability and accelerating diagnosis of runtime and data issues.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*
- Developed full-stack React and Flask features handling 10,000+ daily requests, translating product needs into maintainable APIs and document-heavy interfaces while improving usability, testing, and release confidence.
- Implemented secure batch-processing workflows for PDF, DOCX, and ZIP ingestion with AWS-backed persistence, reducing feedback turnaround by **60%** while supporting stable client-facing production operations.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*
- Built Python-based evaluation pipelines across 100,000+ records, creating reproducible reporting and debugging artifacts that improved system understanding, implementation quality, and technical communication.
- Produced technical reports and targeted validation scenarios translating findings into actionable guidance, strengthening design-review discussions, reducing handoff ambiguity, and supporting better engineering decisions across teams.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations, structured delivery, and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving release confidence across collaborative engineering teams.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular service boundaries and responsive user-facing interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Frontend: TypeScript, JavaScript, React, Next.js, Angular familiarity, HTML/CSS, workflow-driven UI development
Backend & APIs: Node.js, Flask, FastAPI, REST APIs, microservices, Ruby on Rails familiarity, Scala familiarity
Data & Platforms: PostgreSQL, MySQL, SQL Server, MongoDB, document workflows, relational schema design, query tuning
Delivery: Git/GitHub, Docker, Kubernetes, AWS, Azure, Jenkins, CI/CD, code reviews, product/design collaboration

Projects

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS web platform for image editing and search, integrating scalable storage, responsive TypeScript interfaces, and premium billing while improving conversion by **18%**.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a secure document platform with semantic search, summarization, and collaborative Q&A, demonstrating document-heavy workflows over 50K+ records while reducing analysis time by **60%**.

FinNexus Code | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a secure fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, scalable data handling, and client-facing workflows for investment-style applications.